

January 7, 2020

Mr. Doug Langworthy
City of Cadillac WWTP
200 N. Lake Street
Cadillac, Michigan 49601



Reference: 0501867.0106

Subject: Whole Effluent Toxicity Test Results

Dear Doug,

Enclosed please find the final results of the following Chronic Toxicity Tests performed on samples of the City of Cadillac WWTP Outfall 001A effluent.

- 17 December 2019, Chronic *Ceriodaphnia dubia* Toxicity Test
- 17 December 2019, Chronic *Pimephales promelas* Toxicity Test

If you have any questions concerning this report or if I can be of any further assistance to you, please feel free to contact me at (616) 738-7308 or via e-mail at bruce.rabe@erm.com.

Yours sincerely,

Bruce A. Rabe
Director, Aquatic Toxicology Laboratory

BAR:km

Enclosures: Whole Effluent Toxicity Test Report

cc: Cindy Tomaszewski
File

FINAL REPORT

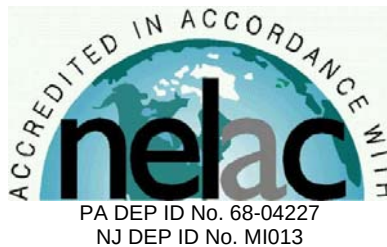
Chronic Toxicity Test
Freshwater Invertebrate,
Ceriodaphnia dubia
EPA Test Method 1002.0

Submitted To:
City of Cadillac WWTP
200 N. Lake Street
Cadillac, MI 49601

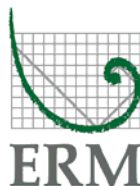
Sample: Outfall 001A

Testing Period: 17 – 24 December 2019

Laboratory I.D. Number: 121719-5



Conducted By:
Environmental Resources Management, Inc.
3352 128th Avenue
Holland, Michigan 49424



Test Overview



Permittee: City of Cadillac WWTP
Location: 1121 Plett Rd.
Cadillac, MI 49601
Contact: Mr. Doug Langworthy
Telephone #: 231.878.4437

NPDES Permit #: MI0020257
Permit Requirements: Acute Toxicity = Report Only
Chronic Toxicity Limit = 1.1 TUc

Test Sample: Outfall 001A
Receiving Water: Clam River

Testing Date: 17 – 24 December 2019

Sample Date(s): 16 December 2019
18 December 2019
20 December 2019

Test/Method: Daphnid, *Ceriodaphnia dubia*,
Survival and Reproduction Test
EPA 821-R-02-013 Method
1002.0.

QC Objectives: Test data met all test
acceptability criteria, except
where noted below.

Data Qualifiers: None

DATA SUMMARY

Effluent Concentrations (%)	Survival (%)	Reproduction (Average Young/Female)
Control	100	33.3
13	100	32.6
25	100	33.4
50	100	29.9
76	100	35.5
100	90	32.2

TEST RESULTS

48-Hour LC ₅₀	>100%
NOEC	100%
LOEC	>100%
ChV	>100%
MSDp	17.3%
Acute Toxicity Unit (TUa)	0
Chronic Toxicity Unit (TUc)	0

TEST CONCLUSION

In accordance with the NPDES permit requirement for the City of Cadillac WWTP, this toxicity test did not exhibit either acute or chronic toxicity.

Bruce A. Rabe
Director, Aquatic
Toxicology Laboratory
ERM Project No. 0501867.0106

Environmental Resources Management
3352 128th Avenue
Holland, Michigan 49424-9263
Phone: 616.399.3500
Fax: 616.399.3777



ERM Testing Method

Ceriodaphnia dubia – Survival and Reproduction Toxicity Test



Upon sample receipt, each effluent sample was analyzed for a suite of water quality parameters (Appendix A - Table 1). Where indigenous organisms were present, the sample was filtered through a 60 micron (µm) NITEX® screen. All samples were maintained at 0 – 6 degrees Celsius (°C) until needed for testing.

A series of five effluent concentrations and a control solution were established for testing. All test solutions were prepared by mixing appropriate volumes of dilution water and effluent in the test containers. Dilution water consisted of reconstituted moderately hard water. The control solution consisted of 100 percent dilution water.

Ceriodaphnia dubia used to initiate this test were obtained from individual, in-house cultures and were less than 24-hours old, and had an age range of 0 to 8 hours at test initiation. Test organisms used to initiate this test were released from adults which met acceptable performance criteria (i.e., ≥15 young/surviving female within 3 broods and obtained from a brood of at least 8 young) and were maintained in reconstituted moderately hard water prior to test initiation.

The *Ceriodaphnia dubia* test was conducted using 30-milliliter (mL) disposable polystyrene containers containing 15 mL of control water or test solution. One *Ceriodaphnia dubia* was added to each test chamber with ten replicate chambers per treatment. Each *Ceriodaphnia dubia* test chamber was fed a 0.2-mL suspension consisting of yeast-Cerophyll-trout chow (YCT) and green algae (*Raphidocelis subcapitata*) mixture daily.

The test solutions were renewed daily during the exposure by transferring the adult daphnid, by way of a wide bore pipette, into fresh control water or test solution.

Percent survival of exposed *Ceriodaphnia dubia* was determined by inspecting for adult mortality daily. Mortality was defined as no body or appendage movement after gentle prodding. Production of young was also determined by daily inspections and enumeration. When 60 percent of the surviving females in the control treatment produced three broods, mean reproduction was determined by calculating the average number of live young produced per female for each treatment.

The test was conducted at a temperature of 25 ± 1°C under fluorescent lighting with a photoperiod of 16 hours light and 8 hours dark. Water quality measurements were performed on all control and test solutions prior to test initiation and on selected treatments daily thereafter, as indicated in the raw data (Appendix A - Table 2).

Following termination of the chronic toxicity test, No Observed Effect Concentrations (NOEC) and Lowest Observed Effect Concentrations (LOEC) were determined for *Ceriodaphnia dubia* survival and reproduction. An NOEC is defined as the highest effluent concentration that does not produce any observed adverse effect to the exposed test organism. An LOEC is defined as the lowest effluent concentration that does produce an observed adverse effect to the exposed test organism. An adverse effect is determined as a statistically significant difference between the control and a given effluent concentration. Significant differences in *Ceriodaphnia dubia* survival were determined using the Fisher's Exact Test.

Prior to the determination of any significant differences in *Ceriodaphnia dubia* reproduction, the data were evaluated for normal distribution and homogeneity characteristics. Depending on the result and the number of test replicates per concentration, an analysis of variance test was performed followed by one of the following mean comparison tests: Dunnett's Procedure, Bonferroni t-Test, Steel's Many-One Rank Test, Wilcoxon Rank Sum Test, or the T-Test. Based on the NOEC and LOEC determinations, a chronic value (ChV) was calculated for the most sensitive of the two test endpoints (i.e., survival or reproduction) and converted to a chronic toxic unit (TUc) for reporting purposes by $100/\text{ChV}$, unless the LOEC was greater than 100 percent in which case the TUc was reported as 0.

To evaluate acute toxicity, a 48-hour LC_{50} and corresponding 95 percent confidence interval was also calculated, where possible. The LC_{50} value estimate was determined by using one of the following statistical methods: graphical, Spearman-Kärber, Trimmed Spearman-Kärber, or Probit. The method selected for reporting test results was determined by the characteristics of the data; that is, the presence or absence of 0 and 100 percent mortality and the number of concentrations in which mortalities between 0 and 100 percent occurred. For reporting purposes, the 48-hour LC_{50} value was converted to an acute toxic unit (TUa) by $100/\text{LC}_{50}$.

For 48-hour LC_{50} values greater than 100 percent, the TUa was derived as follows: (1) if mortality in the 100 percent effluent was 0 percent to 10 percent, the TUa was reported as 0, (2) if mortality in the 100 percent effluent was 11 percent to 49 percent, the TUa was calculated and reported as 0.02 times the percent mortality in 100 percent effluent. All statistical analyses were performed using the CETIS™ Version 1.9.4.3 software program.

The reference toxicant, sodium chloride, was used to monitor the sensitivity of the test organisms and the precision of the testing procedure. Chronic reference toxicant tests are performed at least monthly and the resulting Inhibition Concentrations (IC_{25}) are plotted to determine if the results are within prescribed limits (Appendix A - Standard Reference Toxicant Data). If the IC_{25} of a particular reference toxicant test does not fall within the expected range of $\pm$ two standard deviations from the mean for a given test organism, the sensitivity of that organism and the overall credibility of the test system is suspect.

Reference:

USEPA. 2002. Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, 4th Ed. U.S. Environmental Protection Agency, Office of Water, Washington, D.C., EPA-821-R-02-013.

Case Narrative



1.0 TEST PERFORMANCE CRITERIA

The quality control results achieved laboratory specifications.

2.0 MODIFICATIONS TO ERM'S STANDARD TEST METHOD

Test was performed in accordance with ERM's standard test method (see page 3).

Appendix A
Supporting Documents

- *Raw Test Data*
- *Statistical Analysis (if necessary)*
- *Chain-of-Custody Forms*
- *Standard Reference Toxicant Data*

**Environmental
Resources
Management**

***Ceriodaphnia dubia* - Chronic Toxicity Test
Initial Water Quality and Test Solution Preparation**

Table 1
Page 1 of 1

Permittee/Client:	City of Cadillac WWTP	Control/Dilution Water:	RMHW
Effluent/Location:	Outfall 001A	Organism Batch #:	235-19
Lab I.D.#:	121719-5	Organism Age:	8-16 hours
Beginning Date:	12/17/19	QC Review:	KM
Ending Date:	12/24/19	QC Review Date:	12/31/19
	Time: 1400		
	Time: 1330		

Initial Water Quality:

Parameter	Units	Effluent			Synthetic Water		
Sample #	--	1	2	3	--	--	--
Lab I.D.#/ Batch #	--	121719-5	121919-2	122119-1	144-19	--	--
Temperature	° C	1	2	1	--	--	--
Dissolved Oxygen	mg / L	9.6	9.3	12.1	--	--	--
pH	S.U.	6.8	7.0	6.9	7.8	--	--
Conductivity	umhos/cm	910	1045	1067	315	--	--
Alkalinity	mg / L CaCO ₃	52	74	100	60	--	--
Hardness	mg / L CaCO ₃	180	200	200	82	--	--
Total Ammonia	mg / L NH ₃	0.14	0.16	0.12	--	--	--
Total Residual Chlorine	mg / L Cl ₂	<0.01	<0.01	<0.01	<0.01	--	--
Total mls of Sodium Thiosulfate added per liter	mg / L	--	--	--	--	--	--
Initials	--	KM	KM	KM	KM	--	--

Treatment (% Effluent)	Effluent (mL)	Dilution (mL)	Test Day	Initials	Effluent Sample #	Synthetic Batch #
Control	0	1200	0	KM	1	144-19
13%	156	1044	1	KM/TMB	1	144-19
25%	300	900	2	SRR	2	144-19
50%	600	600	3	TMB/SRR	2	144-19
76%	912	288	4	KM	3	144-19
100%	1200	0	5	KM	3	144-19
			6	MS/KM	3	144-19
			7	SRR	--	--

**Environmental
Resources
Management**

**Ceriodaphnia dubia - Chronic Toxicity Test
Water Quality Data**

Table 2
Page 1 of 1

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 121719-5

Water Quality Data:

Dissolved Oxygen (mg/L)														
Day														
Meter #	3	3	3	3	3	3	3	3	3	3	3	3	5	3
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	8.5	8.1	8.4	8.1	8.5	8.3	8.5	8.2	8.5	8.1	8.4	8.2	8.4	8.1
13%	8.5	8.1	8.4	8.1	8.5	8.3	8.5	8.2	8.5	8.1	8.4	8.2	8.4	8.1
25%	8.5	8.0	8.4	8.2	8.5	8.4	8.5	8.3	8.5	8.2	8.5	8.2	8.4	8.1
50%	8.6	8.0	8.5	8.2	8.4	8.3	8.6	8.3	8.6	8.2	8.6	8.2	8.4	8.1
76%	8.7	8.0	8.5	8.1	8.3	8.2	8.6	8.2	8.6	8.1	8.6	8.1	8.5	8.1
100%	8.7	8.6	8.6	8.1	8.2	8.3	8.6	8.2	8.6	8.1	8.6	8.1	8.5	8.1
pH (S.U.)														
Day														
Meter #	10	10	10	10	10	9	10	10	10	10	10	10	8	10
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	7.8	7.7	7.8	7.7	7.8	7.6	7.8	7.7	7.8	7.8	7.8	7.7	7.8	7.7
13%	--	7.7	--	7.7	--	7.7	--	7.7	--	7.8	--	7.7	--	7.8
25%	--	7.7	--	7.8	--	7.7	--	7.8	--	7.8	--	7.8	--	7.8
50%	--	7.7	--	7.8	--	7.7	--	7.8	--	7.9	--	7.8	--	7.8
76%	--	7.6	--	7.7	--	7.8	--	7.8	--	7.9	--	7.9	--	7.9
100%	6.9	7.6	6.9	7.7	7.1	7.8	7.0	7.7	7.1	8.0	7.1	8.0	7.1	7.9
Conductivity (umhos / cm)														
Day														
Meter #	4	--	4	--	4	--	4	--	4	--	4	--	3	--
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	316	--	315	--	313	--	317	--	316	--	316	--	320	--
13%	388	--	389	--	397	--	399	--	404	--	401	--	411	--
25%	456	--	459	--	475	--	477	--	490	--	488	--	501	--
50%	600	--	602	--	635	--	637	--	660	--	658	--	680	--
76%	770	--	773	--	821	--	830	--	844	--	845	--	859	--
100%	886	--	891	--	975	--	988	--	1026	--	1022	--	1025	--
Temperature (° C)														
Day														
Meter #	3	3	3	3	3	3	3	3	3	3	3	3	5	3
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	24	24	24	24	24	24	24	24	24	24	24	24	24	24
13%	24	24	24	24	24	24	24	24	24	24	24	24	24	24
25%	24	24	24	24	25	24	24	24	24	24	24	24	24	24
50%	24	24	24	24	25	24	24	24	25	24	24	24	24	24
76%	24	24	24	24	26	24	24	24	25	24	24	24	24	24
100%	24	24	24	24	26	24	24	24	25	24	24	24	24	24
I = Initial Chemistry F = Final Chemistry														

Note: D.O. meter also used for temperature measurement unless otherwise noted.

**Environmental
Resources
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***Ceriodaphnia dubia* - Chronic Toxicity Test
Survival and Reproduction Data**

Table 3
Page 1 of 2

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 121719-5

Treatment (% Effluent)	Day No.	Replicate										Average Young/ Female	Number of Live Adults (% Sur.)	Average Young/ Female % CV
		1	2	3	4	5	6	7	8	9	10			
Control	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	--	--	--	--	--	--	--	--	--	--		10	
	4	7	6	6	4	6	5	5	6	7	8		10	
	5	13	11	18	13	11	16	10	13	11	12		10	
	6	--	--	--	--	--	--	--	--	18	--		10	
	7	12	11	14	18	16	19	8	9	--	19		10	
Totals:		32	28	36	35	33	40	24	26	36	34	33.3	(100)	16.0
# Broods (% 3rd Brood)		3	3	3	3	3	3	3	3	3	3	(100)		
13%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	--	--	--	--	--	--	--	--	--	--		10	
	4	6	6	6	5	7	5	6	7	7	7		10	
	5	12	12	15	14	11	14	12	13	12	11		10	
	6	--	--	--	19	--	17	--	--	20	--		10	
	7	17	12	17	--	17	--	1	2	--	16		10	
Totals:		35	30	36	38	35	36	19	22	39	34	32.6	(100)	21.2
25%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	--	--	--	--	--	--	--	--	--	--		10	
	4	6	8	7	6	6	5	6	5	6	7		10	
	5	--	14	15	13	14	15	14	11	13	13		10	
	6	12	--	14	--	16	13	--	14	18	16		10	
	7	17	8	--	12	--	--	10	--	--	--		10	
Totals:		35	30	36	31	36	33	30	30	37	36	33.4	(100)	8.7
50%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	--	--	--	--	--	--	--	--	--	--		10	
	4	8	8	6	6	7	5	6	7	6	5		10	
	5	14	10	16	14	11	16	14	9	12	11		10	
	6	--	--	14	--	16	--	--	--	--	8(S)		10	
	7	11	8	--	9	--	10	8	4	13	7(S)		10	
Totals:		33	26	36	29	34	31	28	20	31	31	29.9	(100)	15.2
76%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	--	--	--	--	--	--	--	--	--	--		10	
	4	7	4	6	8	7	7	6	6	7	6		10	
	5	16	11	13	14	12	13	15	(E)	15	10		10	
	6	--	--	17	20	--	--	12	--	20	--		10	
	7	19	20	--	--	19	20	14	X	--	11		10	
Totals:		42	35	36	42	38	40	35	18	42	27	35.5	(100)	21.7
100%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	--	--	--	--	--	--	--	--	--	--		10	
	4	7	3(S)	6	6	7	5	5	6	6	6		10	
	5	14	4(S)	15	12	11	12	9	13	14	11		10	
	6	--	12	--	--	--	--	14X	8(S)	17	--		9	
	7	18	16	12	16	15	12	X	8(S)	--	4(S)		9	
Totals:		39	35	33	34	33	29	28	33	37	21	32.2	(90)	15.9

X = DEAD ADULT

1X = DEAD ADULT, ONE YOUNG PRODUCED BEFORE DEATH

-- = NO YOUNG RECORDED

(E) = ABORTED EMBRYOS /EGGS

(1) = ONE DEAD YOUNG

(S) = SPLIT BROOD

* = 4th BROOD EXCLUDED FROM TOTAL

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 121719-5

Brood Board Information:

Replicate	1	2	3	4	5	6	7	8	9	10	Brood Board Date: <i>See Below</i>
Chamber Number	<i>5</i>	<i>6</i>	<i>7</i>	<i>17</i>	<i>27</i>	<i>37</i>	<i>48</i>	<i>58</i>	<i>15</i>	<i>55</i>	Young Age Range: <i>8 - 16</i> hours
	<i>12/04/19</i>							<i>12/11/19</i>			

Test Information:

	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
YCT Batch #:	25-19	25-19	25-19	25-19	25-19	25-19	26-19	—
Algae Batch #:	26-19	26-19	26-19	26-19	27-19	27-19	27-19	—
Observation Time:	1400	1130	1330	1300	1600	1300	1300	1330
Initials:	KM	KM	KM	MB/SR	KM	KM	KM	SR
Date:	12/17/19	12/18/19	12/19/19	12/20/19	12/21/19	12/22/19	12/23/19	12/24/19

Comment Section:

[illegible]

CETIS Analytical Report

Report Date: 30 Dec-19 11:36 (p 1 of 2)
Test Code/ID: 354D8028 / 08-9427-1528

Ceriodaphnia 7-d Survival and Reproduction Test

ERM

Analysis ID: 07-8415-6185	Endpoint: 7d Survival Rate	CETIS Version: CETISv1.9.4
Analyzed: 30 Dec-19 11:35	Analysis: STP 2xK Contingency Tables	Status Level: 1
Batch ID: 16-6087-0917	Test Type: Reproduction-Survival (7d)	Analyst: Lab Tech
Start Date: 17 Dec-19 14:00	Protocol: EPA/821/R-02-013 (2002)	Diluent: Reconstituted Water
Ending Date: 24 Dec-19 13:30	Species: Ceriodaphnia dubia	Brine:
Test Length: 6d 23h	Taxon: Branchiopoda	Source: In-House Culture Age: <24
Sample ID: 09-0065-6530	Code: 35AEED92	Project: WET Compliance Testing
Sample Date: 16 Dec-19 07:00	Material: POTW Effluent	Source: City of Cadillac WWTP
Receipt Date: 17 Dec-19 11:30	CAS (PC):	Station: Outfall 001A
Sample Age: 31h (1 °C)	Client: City of Cadillac WWTP	

Data Transform	Alt Hyp	NOEL	LOEL	TOEL	TU
Untransformed	C > T	100	>100	n/a	1

Fisher Exact/Bonferroni-Holm Test

Control	vs	Group	Test Stat	P-Type	P-Value	Decision(α:5%)
Lab Water		13	1.0000	Exact	1.0000	Non-Significant Effect
		25	1.0000	Exact	1.0000	Non-Significant Effect
		50	1.0000	Exact	1.0000	Non-Significant Effect
		76	1.0000	Exact	1.0000	Non-Significant Effect
		100	0.5000	Exact	1.0000	Non-Significant Effect

Test Acceptability Criteria

TAC Limits

Attribute	Test Stat	Lower	Upper	Overlap	Decision
Control Resp	1	0.8	>>	Yes	Passes Criteria

Data Summary

Conc-%	Code	NR	R	NR + R	Prop NR	Prop R	%Effect
0	L	10	0	10	1	0	0.0%
13		10	0	10	1	0	0.0%
25		10	0	10	1	0	0.0%
50		10	0	10	1	0	0.0%
76		10	0	10	1	0	0.0%
100		9	1	10	0.9	0.1	10.0%

7d Survival Rate Detail

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4	Rep 5	Rep 6	Rep 7	Rep 8	Rep 9	Rep 10
0	L	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
13		1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
25		1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
50		1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
76		1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
100		1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000	1.0000	1.0000

7d Survival Rate Binomials

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4	Rep 5	Rep 6	Rep 7	Rep 8	Rep 9	Rep 10
0	L	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1
13		1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1
25		1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1
50		1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1
76		1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1
100		1/1	1/1	1/1	1/1	1/1	1/1	0/1	1/1	1/1	1/1

CETIS Analytical Report

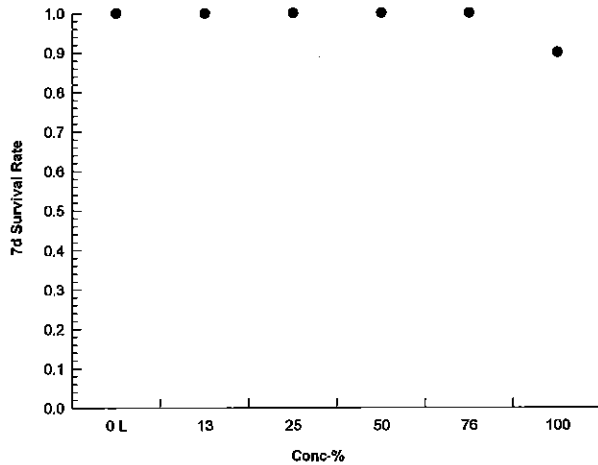
Report Date: 30 Dec-19 11:36 (p 2 of 2)
Test Code/ID: 354D8028 / 08-9427-1528

Ceriodaphnia 7-d Survival and Reproduction Test

ERM

Analysis ID: 07-8415-6185	Endpoint: 7d Survival Rate	CETIS Version: CETISv1.9.4
Analyzed: 30 Dec-19 11:35	Analysis: STP 2xK Contingency Tables	Status Level: 1

Graphics



CETIS Analytical Report

Report Date: 30 Dec-19 11:36 (p 1 of 2)
Test Code/ID: 354D8028 / 08-9427-1528

Ceriodaphnia 7-d Survival and Reproduction Test

ERM

Analysis ID: 03-7946-5552	Endpoint: Reproduction	CETIS Version: CETISv1.9.4
Analyzed: 30 Dec-19 11:35	Analysis: Nonparametric-Control vs Treatments	Status Level: 1
Batch ID: 16-6087-0917	Test Type: Reproduction-Survival (7d)	Analyst: Lab Tech
Start Date: 17 Dec-19 14:00	Protocol: EPA/821/R-02-013 (2002)	Diluent: Reconstituted Water
Ending Date: 24 Dec-19 13:30	Species: Ceriodaphnia dubia	Brine:
Test Length: 6d 23h	Taxon: Branchiopoda	Source: In-House Culture Age: <24
Sample ID: 09-0065-6530	Code: 35AEED92	Project: WET Compliance Testing
Sample Date: 16 Dec-19 07:00	Material: POTW Effluent	Source: City of Cadillac WWTP
Receipt Date: 17 Dec-19 11:30	CAS (PC):	Station: Outfall 001A
Sample Age: 31h (1 °C)	Client: City of Cadillac WWTP	

Data Transform	Alt Hyp	NOEL	LOEL	TOEL	TU	PMSD
Untransformed	C > T	100	>100	n/a	10	17.31%

Steel Many-One Rank Sum Test

Per M06666

Control	vs	Conc-%	Test Stat	Critical	Ties	DF	P-Type	P-Value	Decision(α:5%)
Lab Water		13	104	75	4	18	Asymp	0.8098	Non-Significant Effect
		25	103.5	75	3	18	Asymp	0.7973	Non-Significant Effect
		50	86	75	3	18	Asymp	0.2336	Non-Significant Effect
		76	120.5	75	4	18	Asymp	0.9913	Non-Significant Effect
		100	99.5	75	4	18	Asymp	0.6816	Non-Significant Effect

Test Acceptability Criteria

TAC Limits

Attribute	Test Stat	Lower	Upper	Overlap	Decision
Control Resp	33.3	15	>>	Yes	Passes Criteria

ANOVA Table

Source	Sum Squares	Mean Square	DF	F Stat	P-Value	Decision(α:5%)
Between	167.083	33.4167	5	1.054	0.3960	Non-Significant Effect
Error	1711.9	31.7019	54			
Total	1878.98		59			

Distributional Tests

Attribute	Test	Test Stat	Critical	P-Value	Decision(α:1%)
Variances	Bartlett Equality of Variance Test	8.871	15.09	0.1143	Equal Variances
Distribution	Shapiro-Wilk W Normality Test	0.9061	0.9459	2.2E-04	Non-Normal Distribution

Reproduction Summary

Conc-%	Code	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	L	10	33.3	29.5	37.1	34	24	40	1.68	15.96%	0.00%
13		10	32.6	27.66	37.54	35	19	39	2.182	21.16%	2.10%
25		10	33.4	31.32	35.48	34	30	37	0.9214	8.72%	-0.30%
50		10	29.9	26.66	33.14	31	20	36	1.433	15.16%	10.21%
76		10	35.5	30	41	37	18	42	2.432	21.67%	-6.61%
100		10	32.2	28.54	35.86	33	21	39	1.618	15.89%	3.30%

Reproduction Detail

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4	Rep 5	Rep 6	Rep 7	Rep 8	Rep 9	Rep 10
0	L	32	28	38	35	33	40	24	28	36	39
13		35	30	38	38	35	36	19	22	39	34
25		35	30	36	31	36	33	30	30	37	36
50		33	26	36	29	34	31	28	20	31	31
76		42	35	36	42	38	40	35	18	42	27
100		39	35	33	34	33	29	28	33	37	21

CETIS Analytical Report

Report Date: 30 Dec-19 11:36 (p 2 of 2)
Test Code/ID: 354D8028 / 08-9427-1528

Ceriodaphnia 7-d Survival and Reproduction Test

ERM

Analysis ID: 03-7946-5552

Endpoint: Reproduction

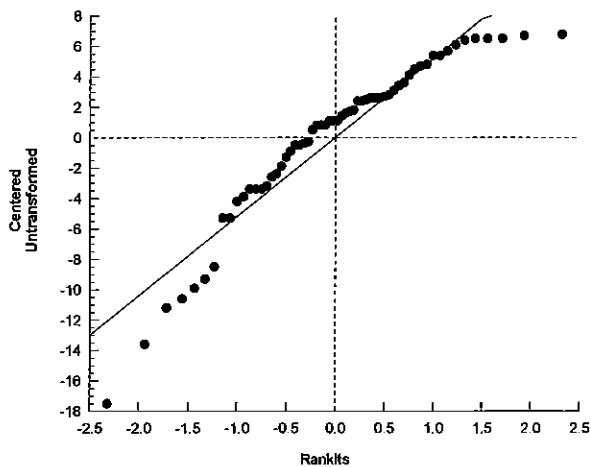
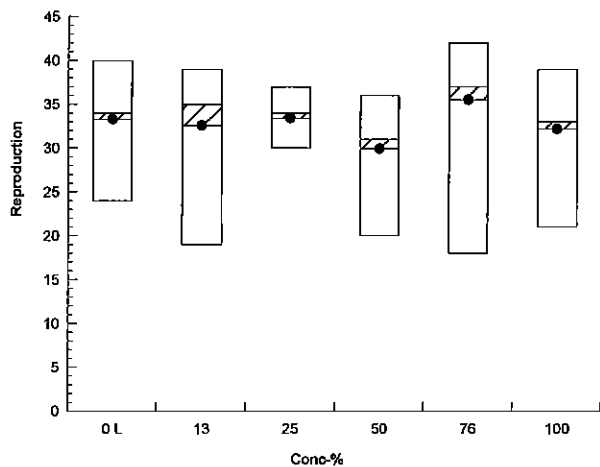
CETIS Version: CETISv1.9.4

Analyzed: 30 Dec-19 11:35

Analysis: Nonparametric-Control vs Treatments

Status Level: 1

Graphics





AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *



ERM

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

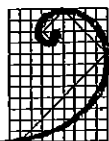
CLIENT NAME:	City of Cadillac			SAMPLER	TL
ADDRESS:	1121 Platt Rd Cadillac, MI			PHONE NUMBER:	231-775-2841
SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS
Tert.	12-17-19	7:00 AM	Comp	3 gal.	pH=6.77 s.u. NH ₃ = mg/L
	12-18-19	7:00 AM			pH= s.u. NH ₃ = mg/L
					pH= s.u. NH ₃ = mg/L
					pH= s.u. NH ₃ = mg/L
					pH= s.u. NH ₃ = mg/L
					pH= s.u. NH ₃ = mg/L
ANALYSES REQUESTED [check item(s)]	Test Material: ☒ Water/Wastewater ☐ Sediment ☐ Product	Test Type: ☒ Acute ☒ Chronic ☐ Other	Test Species: ☒ Ceriodaphnia dubia ☐ Daphnia magna ☐ Daphnia pulex ☒ Fathead minnow (Pimephales promelas)		
COMMENT SECTION: ONLY CHRONIC - ON TWO SPECIES. ON 12/10/19					

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
T. Saepko / City of Cadillac	12-18-19		Tonyon Bragg / ERM	12/19/19	1100

* See Instructions for Sample Collection on Back of Sheet

February 2018



ERM®

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	CITY OF CADILLAC				SAMPLER	CHAD M.				
ADDRESS:	1121 PLETT RD. CADILLAC				PHONE NUMBER:	231-775-2841				
SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin/End)	TIME (Begin/End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
	12-19	7:00am	C	4-1 gal.	pH= 7.05 s.u. NH ₃ = .24 mg/L	122119-1	Temp. (°C) ☒ On Ice	D.O. mg/L 12	pH s.u. 6.9	Cond umhos/cm 1067
	12-20	7:00am			pH= s.u. NH ₃ = mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm

ANALYSES REQUESTED [check item(s)]	Test Material: ☒ Water/Wastewater ☐ Sediment ☐ Product	Test Type: ☐ Acute ☒ Chronic ☐ Other	Test Species: ☒ Ceriodaphnia dubia ☐ Daphnia magna ☐ Daphnia pulex ☒ Fathead minnow (Pimephales promelas) ☐ Rainbow Trout (Oncorhynchus mykiss) ☐ Sheephead minnow (Cyprinodon variegatus) ☐ Silverside minnow (Menidia beryllina) ☐ Other (write in comments section)
---------------------------------------	---	---	--

COMMENT SECTION:			
------------------	--	--	--

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
Chad Waveguard	12-20-19	1:30pm	Ship Next Day / Sat WPS Katie Mardian / ERM	12/21/19	1130

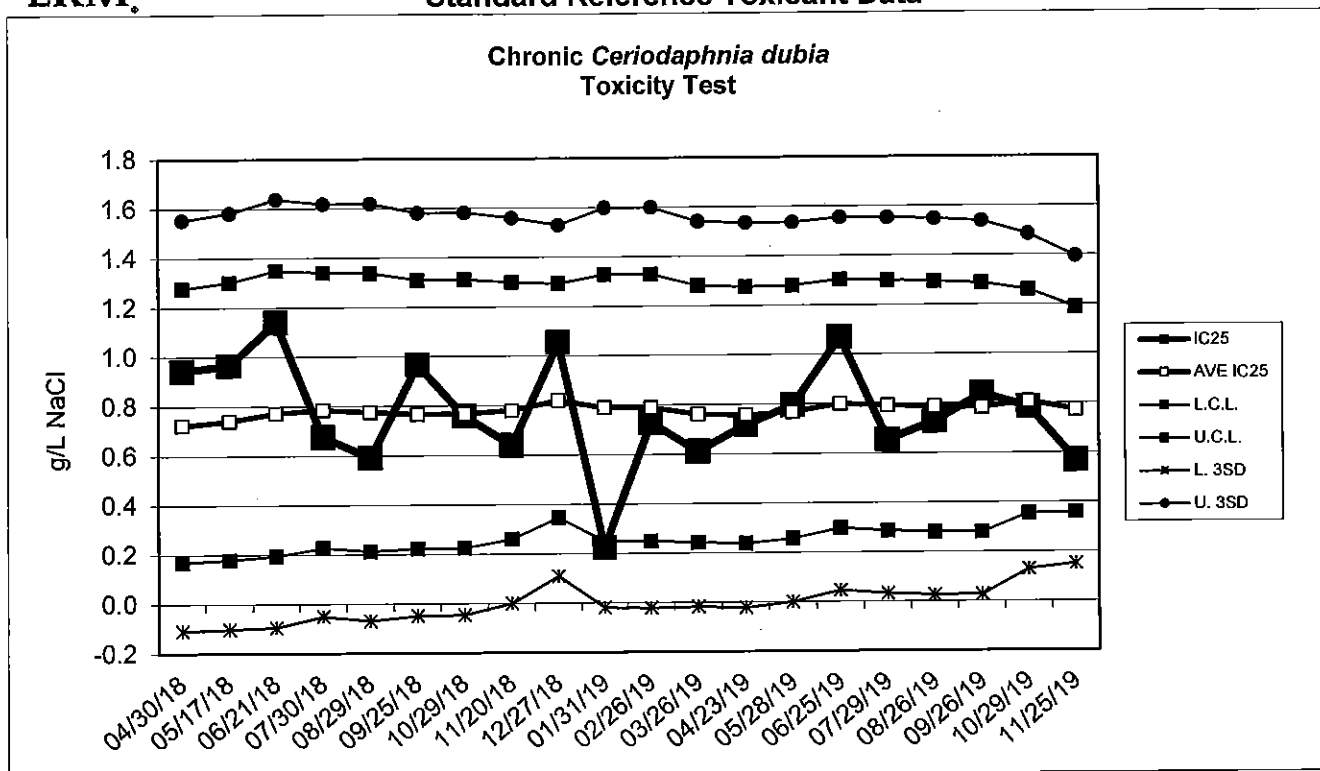
* See Instructions for Sample Collection on Back of Sheet

February 2018



Environmental Resources Management

Standard Reference Toxicant Data



Chronic *Ceriodaphnia dubia* Toxicity Test Data

Date	IC25 (g/L NaCl)	AVE IC25 (g/L NaCl)	CONTROL LIMIT		Survival (%)	CONTROL Reproduction (ave. young)	CV (%)
			Lower	Upper			
04/30/18	0.94	0.7	0.17	1.28	100	32.0	25.5
05/17/18	0.97	0.7	0.18	1.30	100	30.0	38.6
06/21/18	1.14	0.8	0.19	1.35	80	35.2	8.2
07/30/18	0.68	0.8	0.23	1.34	100	25.5	16.3
08/29/18	0.59	0.8	0.21	1.34	100	30.1	26.2
09/25/18	0.97	0.8	0.22	1.31	100	27.6	26.7
10/29/18	0.76	0.8	0.22	1.31	100	32.7	24.8
11/20/18	0.64	0.8	0.26	1.30	100	34.8	15.2
12/27/18	1.06	0.8	0.35	1.29	100	26.8	43.7
01/31/19	0.23	0.8	0.25	1.33	100	34.7	14.9
02/26/19	0.73	0.8	0.25	1.33	100	27.9	9.3
03/26/19	0.61	0.8	0.24	1.28	100	40.2	9.9
04/23/19	0.72	0.8	0.24	1.28	100	36.1	25.4
05/28/19	0.79	0.8	0.26	1.28	100	37.6	3.1
06/25/19	1.07	0.8	0.30	1.30	100	29.4	26.7
07/29/19	0.65	0.8	0.29	1.30	100	33.7	14.6
08/26/19	0.73	0.8	0.28	1.29	100	30.4	23.5
09/26/19	0.84	0.8	0.28	1.29	100	24.8	27.4
10/29/19	0.79	0.8	0.35	1.26	100	37.1	15.9
11/25/19	0.57	0.8	0.36	1.19	100	28.4	30.7

Appendix B
MDEGLE Reporting Form



MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

CERIODAPHNIA DUBIA CHRONIC TOXICITY TEST REPORT

By authority of PA 451 of 1994, as amended.

INSTRUCTIONS: Use this form to report chronic toxicity test results. Use separate forms for more than 1 test.

1. NAME OF FACILITY (on NPDES permit) City of Cadillac WWTP				2. NPDES PERMIT # M I 0 0 2 0 2 5 7							
1. RECEIVING WATER (as designated in permit) Clam River				2. OUTFALL 001A				5. RECEIVING WATER CONCENTRATION (if known)			
6. TEST LAB (Name and Address) Environmental Resources Management 3352 128 th Avenue Holland, MI 49424-9263											
7. TEST START DATE 12/17/19			8. TEST END DATE 12/24/19			9. AGE RANGE OF ORGANISMS AT TEST START 8 – 16 Hours Old			10. REPORT DATE 01/07/20		
11. NAME OF PERSON CONDUCTING TEST Lab Contact: Mark Baker						12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT Bruce Rabe (616) 738 – 7308					
13. SAMPLE COLLECTION DATES Sample 1: 12/16/19 Sample 2: 12/18/19 Sample 3: 12/20/19				14. DATE RECEIVED Sample 1: 12/17/19 Sample 2: 12/19/19 Sample 3: 12/21/19				15. ARRIVAL TEMP (C°) Sample 1: 1 Sample 2: 2 Sample 3: 1			
16. DATE OF FIRST USE Sample 1: 12/17/19 Sample 2: 12/19/19 Sample 3: 12/21/19				17. TOTAL RESIDUAL CHLORINE (mg/l) Sample 1: <0.01 Sample 2: <0.01 Sample 3: <0.01				18. AMMONIA (mg/l as N) Sample 1: 0.14 Sample 2: 0.18 Sample 3: 0.12			
19. WAS SAMPLE DECHLORINATED? Sample 1: ☐ YES ☒ NO IF YES ➡ Sample 2: ☐ YES ☒ NO IF YES ➡ Sample 3: ☐ YES ☒ NO IF YES ➡				20. DESCRIBE DECHLORINATION (if any)							
21. EFFLUENT SAMPLES WERE COLLECTED (check one) ☐ BEFORE CHLORINATION ☐ AFTER CHLORINATION ☐ AFTER CHLORINATION, BEFORE DECHLORINATION ☐ AFTER DECHLORINATION ☒ FACILITY DOES NOT CHLORINATE											
22. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded holding time.) N/A											
23. EFFLUENT FILTERED? ☐ YES ☒ NO IF YES ➡						24. STATE MESH SIZE OF FILTER					
25. EFFLUENT SAMPLE TYPE (check one type for each sample) Sample 1: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE Sample 2: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE Sample 3: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE								26. IDENTIFY THE DILUENT (O ₁) CONTROL <u>RMHW</u> IDENTIFY THE SECONDARY (O ₂) CONTROL (if used) _____			
27. SUMMARY OF DATA AND RESULTS - SURVIVAL AND REPRODUCTION											
CONCENTRATION OF EFFLUENT (%)		O ₁	O ₂	13%	25%	50%	76%	100%			
48-HOUR SURVIVAL (%)		100	N/A	100	100	100	100	100			
7-DAY MEAN REPRODUCTION/FEMALE		33.3	N/A	32.6	33.4	29.9	35.5	32.2			
7-DAY MEAN SURVIVAL (%)		100	N/A	100	100	100	100	90			
28. 48-HOUR LC ₅₀ (%) >100				29. TU _a (acute toxic units) 0							
30. 7-DAY CHRONIC VALUE (%) >100		31. NOEC 100%		32. LOEC >100%		33. TU _c (chronic toxic units) 0					

EQP 5819 (02/10)

FINAL REPORT

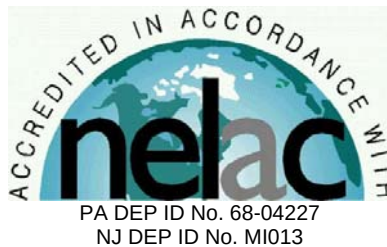
Chronic Toxicity Test
Freshwater Vertebrate,
Pimephales promelas
EPA Test Method 1000.0

Submitted To:
City of Cadillac WWTP
200 N. Lake Street
Cadillac, MI 49601

Sample: Outfall 001A

Testing Period: 17 – 24 December 2019

Laboratory I.D. Number: 121719-5



Conducted By:
Environmental Resources Management, Inc.
3352 128th Avenue
Holland, Michigan 49424



Test Overview



Permittee: City of Cadillac WWTP
Location: 1121 Plett Rd.
Cadillac, MI 49601
Contact: Mr. Doug Langworthy
Telephone #: 231.878.4437

NPDES Permit #: MI0020257
Permit Requirements: Report
Test Sample: Outfall 001A
Receiving Water: Clam River

Testing Date: 17 – 24 December 2019

Sample Date(s): 16 December 2019
18 December 2019
20 December 2019

Test/Method: Fathead Minnow,
Pimephales promelas,
Survival and Growth Test
EPA 821-R-02-013
Method 1000.0.

QC Objectives: Test data met all test
acceptability criteria, except
where noted below.

Data Qualifiers: See Case Narrative.

DATA SUMMARY

Effluent Concentrations (%)	Survival (%)	Growth Average Wt./ Organism (mg)
Control	95	0.491
13	92.5	0.473
25	90	0.504
50	92.5	0.557
76	95	0.548
100	100	0.531

TEST RESULTS

96-Hour LC ₅₀	>100%
NOEC	100%
LOEC	>100%
ChV	>100%
MSDp (Growth)	34.2% ^{DQ}
Acute Toxicity Unit (TUa)	0
Chronic Toxicity Unit (TUC)	0

^{DQ} See Data Qualifiers in Case Narrative

TEST CONCLUSION

In accordance with the NPDES permit requirements for the City of Cadillac WWTP, this toxicity test did not exhibit either acute or chronic toxicity.

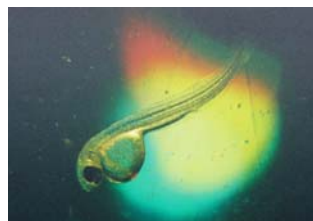
Bruce A. Rabe
Director, Aquatic
Toxicology Laboratory
ERM Project No. 0501867.0106

Environmental Resources Management
3352 128th Avenue
Holland, Michigan 49424-9263
Phone: 616.399.3500
Fax: 616.399.3777



ERM Testing Method

Pimephales promelas – Survival and Growth Toxicity Test



Upon sample receipt, each effluent sample was analyzed for a suite of water quality parameters (Appendix A - Table 1). Where indigenous organisms were present, the sample was filtered through a 60 micron (μm) NITEX® screen. All samples were maintained at 0 – 6 degrees Celsius ($^{\circ}\text{C}$) until needed for testing.

A series of five effluent concentrations and a control solution were established for testing. All test solutions were prepared by mixing appropriate volumes of dilution water and effluent in the test containers. Dilution water consisted of reconstituted moderately hard water. The control solution consisted of 100 percent dilution water.

Pimephales promelas used to initiate this test were obtained from in-house cultures and were less than 24-hours old at test initiation. Test organisms were maintained in reconstituted moderately hard water prior to test initiation.

The *Pimephales promelas* test was conducted using 300 to 500-milliliter (mL) disposable polypropylene containers containing 250 mL of control water or test solution. Ten fish were randomly added to each test chamber with four replicate chambers per treatment. Each *Pimephales promelas* test chamber was fed 0.1 mL of a concentrated suspension of less than 24-hour old live brine shrimp nauplii (*Artemia* sp.) two times per day. Test solutions were renewed daily during the exposure by replacing approximately 90 percent of the 24-hour old solution with fresh control water or appropriate test solution. Prior to test solution renewal, uneaten and dead brine shrimp, along with other debris, were removed from the bottom of the test chambers.

Percent survival of exposed *Pimephales promelas* was determined daily by enumeration of live organisms. Mortality was defined as no body movement after gentle prodding. At the termination of the chronic test, larvae in each test chamber were counted, dried, and weighed to the nearest 0.01 milligram (mg) on an analytical balance.

The test was conducted at a temperature of $25 \pm 1^{\circ}\text{C}$ under fluorescent lighting with a photoperiod of 16 hours light and 8 hours dark. Water quality measurements were performed on all control and test solutions prior to test initiation and on selected treatments daily thereafter, as indicated in the raw data (Appendix A - Table 2).

Following termination of the chronic toxicity test, No Observed Effect Concentrations (NOEC) and Lowest Observed Effect Concentrations (LOEC) were determined for *Pimephales promelas* survival and growth. An NOEC is defined as the highest effluent concentration which does not produce any observed adverse effect to the exposed test organism. An LOEC is defined as the lowest effluent concentration which does produce an observed adverse effect to the exposed test organism. An adverse effect is determined as a statistically significant difference between the control and a given effluent concentration.

Prior to the determination of any significant differences in *Pimephales promelas* survival and growth, the data were evaluated for normal distribution and homogeneity characteristics. Depending on the result and the number of test replicates per concentration, an analysis of variance test was performed, followed by one of the following mean comparison tests: Dunnett's Procedure, Bonferroni t-Test, Steel's Many-One Rank Test, Wilcoxon Rank Sum Test, or the T-Test. Based on the NOEC and LOEC determinations, a chronic value (ChV) was calculated for the most sensitive of the two test endpoints (i.e., survival or growth) and converted to a chronic toxic unit (TUC) for reporting purposes by $100/\text{ChV}$, unless the LOEC was greater than 100 percent in which case the TUC was reported as 0.

To evaluate acute toxicity, a 96-hour LC₅₀ and corresponding 95 percent confidence interval were also calculated, where possible. The LC₅₀ value estimate was determined by using one of the following statistical methods: graphical, Spearman-Kärber, Trimmed Spearman-Kärber, or Probit. The method selected for reporting test results was determined by the characteristics of the data; that is, the presence or absence of 0 and 100 percent mortality and the number of concentrations in which mortalities between 0 and 100 percent occurred. For reporting purposes, the 96-hour LC₅₀ value was converted to an acute toxic unit (TUa) by 100/LC₅₀.

For 96-hour LC₅₀ values greater than 100 percent, the TUa was derived as follows: (1) if mortality in the 100 percent effluent was 0 to 10 percent, the TUa was reported as 0, (2) if mortality in the 100 percent effluent was 11 to 49 percent, the TUa was calculated and reported as 0.02 times the percent mortality in the 100 percent effluent. All statistical analyses were performed using the CETIS™ Version 1.9.4.3 software program

The reference toxicant, sodium chloride, was used to monitor the sensitivity of the test organisms. Chronic reference toxicant tests are performed at least monthly and the resulting Inhibition Concentrations (IC₂₅) are plotted to determine if the results are within prescribed limits (Appendix A - Standard Reference Toxicant Data). If the IC₂₅ of a particular reference toxicant test does not fall within the expected range of $\pm$ two standard deviations from the mean for a given test organism, the sensitivity of that organism and the overall credibility of the test system is suspect.

Reference:

USEPA. 2002. Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, 4th Ed. U.S. Environmental Protection Agency, Office of Water, Washington, D.C., EPA-821-R-02-013.

Case Narrative



1.0 TEST PERFORMANCE CRITERIA

The quality control results achieved laboratory specifications.

2.0 MODIFICATIONS TO ERM'S STANDARD TEST METHOD

Test was performed in accordance with ERM's standard test method (see page 3).

3.0 DATA QUALIFIERS

The percent minimum significant difference (MSDp) value of 34.2 percent for growth exceeded the maximum allowed value of 30 percent. An MSDp exceedance can mask statistically significant reductions in growth when comparing effluent responses to the control. However, based on the greater response for both survival and growth in the 100 percent effluent when compared to the control, the MSDp exceedance was deemed a non-issue.

Appendix A
Supporting Documents

- *Raw Test Data*
- *Statistical Analysis (if necessary)*
- *Chain-of-Custody Forms*
- *Standard Reference Toxicant Data*

**Environmental
Resources
Management**

***Pimephales promelas* - Chronic Toxicity Test
Initial Water Quality and Test Solution Preparation**

Table 1
Page 1 of 1

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 121719-5
Beginning Date: 12/17/19
Ending Date: 12/24/19

Time: 1600
Time: 1500

Control/Dilution Water: RMHW
Organism Batch #: E16-14
Organism Age: 224 hrs
QC Review: KM
QC Review Date: 12/31/19

Initial Water Quality:

Parameter	Units	Effluent			Synthetic Water		
Sample #	--	1	2	3	--	--	--
Lab I.D.#/ Batch #	--	121719-5	121919-2	122119-1	144-19	--	--
Temperature	° C	1	2	1	--	--	--
Dissolved Oxygen	mg / L	9.6	9.3	12.1	--	--	--
pH	S.U.	6.8	7.0	6.9	7.8	--	--
Conductivity	umhos/cm	910	1045	1067	315	--	--
Alkalinity	mg / L CaCO ₃	52	74	100	60	--	--
Hardness	mg / L CaCO ₃	180	200	200	82	--	--
Total Ammonia	mg / L NH ₃	0.14	0.16	0.12	--	--	--
Total Residual Chlorine	mg / L Cl ₂	<0.01	<0.01	<0.01	<0.01	--	--
Total mls of Sodium Thiosulfate added per liter	mg / L	--	--	--	--	--	--
Initials	--	KM	KM	KM	SPR	--	--

Test Solution Preparation: Test Solution Prepared For Both Species.

Treatment (% Effluent)	Effluent (mL)	Dilution (mL)	Test Day	Initials	Effluent Sample #	Synthetic Batch #
Control	0	1200	0	KM	1	144-19
13%	156	1044	1	KM/TMB	1	144-19
25%	300	900	2	SPR	2	144-19
50%	600	600	3	TMB/SPR	2	144-19
76%	912	288	4	KM	3	144-19
100%	1200	0	5	KM	3	144-19
			6	MS/KM	3	144-19
			7	MS	--	--

**Environmental
Resources
Management**

**Pimephales promelas - Chronic Toxicity Test
Water Quality Data**

Table 2
Page 1 of 1

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 121719-5

Water Quality Data:

Dissolved Oxygen (mg/L)														
Day														
Meter #	3	3	3	3	3	3	3	3	3	3	3	3	5	5
Treatment	0	1	2	3	4	5	6	7	8	9	10	11	12	13
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	8.5	5.9	8.4	7.0	8.5	6.0	8.5	7.0	8.5	6.6	8.4	6.4	8.4	7.7
13%	8.5	5.9	8.4	6.9	8.5	6.0	8.5	6.8	8.5	6.7	8.4	5.6	8.4	7.7
25%	8.5	5.8	8.4	6.8	8.5	6.2	8.5	6.6	8.5	6.6	8.5	5.9	8.4	7.7
50%	8.6	5.9	8.5	6.8	8.4	5.8	8.6	6.6	8.6	6.2	8.6	5.9	8.4	7.0
76%	8.7	6.0	8.5	6.6	8.3	5.7	8.6	6.3	8.6	5.8	8.6	5.9	8.5	6.5
100%	8.7	5.9	8.6	6.7	8.2	4.8	8.6	6.2	8.6	5.3	8.6	5.8	8.5	6.1
pH (S.U.)														
Day														
Meter #	10	10	10	10	10	10	10	10	10	10	10	10	8	9
Treatment	0	1	2	3	4	5	6	7	8	9	10	11	12	13
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	7.8	7.6	7.8	7.5	7.6	7.0	7.6	7.3	7.8	7.4	7.8	7.4	7.8	7.5
13%	--	7.6	--	7.5	--	7.2	--	7.2	--	7.5	--	7.3	--	7.5
25%	--	7.5	--	7.5	--	7.2	--	7.3	--	7.5	--	7.3	--	7.6
50%	--	7.5	--	7.4	--	7.2	--	7.2	--	7.4	--	7.4	--	7.6
76%	--	7.5	--	7.4	--	7.3	--	7.2	--	7.4	--	7.4	--	7.8
100%	6.9	7.4	6.9	7.4	7.1	7.1	7.0	7.1	7.1	7.4	7.1	7.4	7.1	7.8
Conductivity (umhos / cm)														
Day														
Meter #	4	--	4	--	4	--	4	--	4	--	4	--	3	--
Treatment	0	1	2	3	4	5	6	7	8	9	10	11	12	13
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	316	--	315	--	313	--	317	--	316	--	316	--	320	--
13%	388	--	389	--	397	--	399	--	404	--	401	--	411	--
25%	456	--	459	--	475	--	477	--	490	--	488	--	501	--
50%	600	--	602	--	635	--	637	--	660	--	658	--	680	--
76%	770	--	773	--	821	--	830	--	844	--	845	--	859	--
100%	886	--	891	--	975	--	989	--	1026	--	1022	--	1025	--
Temperature (° C)														
Day														
Meter #	3	3	3	3	3	3	3	3	3	3	3	3	5	5
Treatment	0	1	2	3	4	5	6	7	8	9	10	11	12	13
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	24	25	24	25	24	24	24	25	24	25	24	25	24	25
13%	24	25	24	25	24	25	24	25	24	25	24	25	24	25
25%	24	25	24	25	25	25	24	25	24	25	24	25	24	25
50%	24	25	24	25	25	25	24	25	23	25	24	25	24	25
76%	24	25	24	25	26	25	24	25	23	25	24	25	24	25
100%	24	25	24	25	26	25	24	23	23	25	24	25	24	25
I = Initial Chemistry F = Final Chemistry														

Note: D.O. meter also used for temperature measurement unless otherwise noted.

**Environmental
Resources
Management**

***Pimephales promelas* - Chronic Toxicity Test
Survival Data**

Table 3
Page 1 of 2

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 121719-5

Survival Data:

Survival Data.

Treatment (% Effluent)	Rep.	# Live Organisms Day								Rep.	# Live Organisms Day								96 Hour Survival Summary		
		0	1	2	3	4	5	6	7		0	1	2	3	4	5	6	7	Total Live		% Survival
																			Initial	Final	
Control	A	10	10	10	10	10	10	9	9	B	10	10	10	9	9	9	9	9	40	39	97.5
13%	A	10	10	10	10	10	10	10	10	B	10	10	10	10	10	10	9	8	40	40	100
25%	A	10	10	10	10	10	10	10	10	B	10	10	8	7	7	7	7	7	40	36	90
50%	A	10	10	10	9	9	9	9	9	B	10	10	10	10	10	10	10	10	40	37	92.5
76%	A	10	10	10	10	10	10	10	10	B	10	10	10	10	10	10	10	10	40	38	95
100%	A	10	10	10	10	10	10	10	10	B	10	10	10	10	10	10	10	10	40	40	100

Treatment (% Effluent)	Rep.	# Live Organisms Day								Rep.	# Live Organisms Day								7 Day Survival Summary		
		0	1	2	3	4	5	6	7		0	1	2	3	4	5	6	7	Total Live		% Survival
																			Initial	Final	
Control	C	10	10	10	10	10	10	10	10	D	10	10	10	10	10	10	10	10	40	38	95
13%	C	10	10	10	9	10	10	10	10	D	10	10	10	10	10	10	9	9	40	37	92.5
25%	C	10	10	10	10	9	9	9	9	D	10	10	10	10	10	10	10	10	40	36	90
50%	C	10	10	8	8	8	8	8	8	D	10	10	10	10	10	10	10	10	40	37	92.5
76%	C	10	10	10	8	8	8	8	8	D	10	10	10	10	10	10	10	10	40	38	95
100%	C	10	10	10	10	10	10	10	10	D	10	10	10	10	10	10	10	10	40	40	100

Test Information:

	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Time:	1100	1400	1400	1400	1330	1430	1430	1500
Initials:	MB	MB/KM	KM	MS/PR	KM	KM	MB/MB	MS
Date:	12/17/19	12/18/19	12/19/19	12/20/19	12/21/19	12/22/19	12/23/19	12/24/19

Feeding:

	Day 0	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Batch #:	350-19	351-19	352-19	353-19	354-19	355-19	356-19	--
Initials AM:	--	MB	MS	MS	KM	KM	KM	--
Initials PM:	MB	MB	MB	PR	KM	KM	KM	--

Oven:

Date In	Time In	Initials	Date Out	Time Out	Initials
12/24/19	1530	MB	12/25/19	1530	MB

Comment Section:

Day	Date	Initials	Comments

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 121719-5

Pan #	Conc. (% Effluent)	Replicate	Final Weight (mg)	Initial Weight (mg)	Larvae Weight (mg)	# of Initial Organisms	Avg. Wt./ Organism/ Replicate (mg)	Avg. Wt./ Organism/ Treatment (mg)	Avg. Wt./ Organism/ Treatment % CV
Date			12/26/19	12/23/19					
Analyst			sb/km	sb/km					
1	Control	A	22.35	17.89	4.46	10	0.446		
2	Control	B	24.31	18.74	5.57	10	0.557		
3	Control	C	22.52	17.68	4.84	10	0.484		
4	Control	D	21.17	16.40	4.77	10	0.477	0.491	9.6
5	13%	A	21.48	16.53	4.95	10	0.495		
6	13%	B	21.49	14.44	7.05	10	0.705		
7	13%	C	19.59	18.00	1.59	10	0.159		
8	13%	D	22.86	17.52	5.34	10	0.534	0.473	48.3
9	25%	A	20.42	15.71	4.71	10	0.471		
10	25%	B	21.50	16.66	4.84	10	0.484		
11	25%	C	21.40	15.98	5.42	10	0.542		
12	25%	D	25.33	20.15	5.18	10	0.518	0.504	6.4
13	50%	A	22.08	16.55	5.53	10	0.553		
14	50%	B	24.77	18.70	6.07	10	0.607		
15	50%	C	25.29	19.74	5.55	10	0.555		
16	50%	D	23.95	18.84	5.11	10	0.511	0.557	7.1
17	76%	A	23.62	18.08	5.54	10	0.554		
18	76%	B	23.86	17.99	5.87	10	0.587		
19	76%	C	22.06	16.73	5.33	10	0.533		
20	76%	D	26.97	21.81	5.16	10	0.516	0.548	5.6
21	100%	A	21.82	16.57	5.25	10	0.525		
22	100%	B	21.84	16.73	5.11	10	0.511		
23	100%	C	24.14	18.65	5.49	10	0.549		
24	100%	D	23.99	18.60	5.39	10	0.539	0.531	3.1

Quality Assurance			Final Wt. (mg)		
25	Blank	A	16.69	16.73	-0.04
26	Blank	B	18.58	18.57	0.01

CETIS Analytical Report

Report Date: 30 Dec-19 11:40 (p 1 of 2)
Test Code/ID: 5B3693B4 / 15-3030-3412

Fathead Minnow 7-d Larval Survival and Growth Test

ERM

Analysis ID: 00-1513-8377	Endpoint: 7d Survival Rate	CETIS Version: CETISv1.9.4
Analyzed: 30 Dec-19 11:40	Analysis: Parametric-Control vs Treatments	Status Level: 1
Batch ID: 10-5585-1802	Test Type: Growth-Survival (7d)	Analyst: Lab Tech
Start Date: 17 Dec-19 16:00	Protocol: EPA/821/R-02-013 (2002)	Diluent: Reconstituted Water
Ending Date: 24 Dec-19 15:00	Species: Pimephales promelas	Brine:
Test Length: 6d 23h	Taxon: Actinopterygii	Source: In-House Culture Age: <24
Sample ID: 06-4191-1844	Code: 2642CC24	Project: WET Compliance Testing
Sample Date: 16 Dec-19 07:00	Material: POTW Effluent	Source: City of Cadillac WWTP
Receipt Date: 17 Dec-19 11:30	CAS (PC):	Station: Outfall 001A
Sample Age: 33h (1 °C)	Client: City of Cadillac WWTP	

Data Transform	Alt Hyp	NOEL	LOEL	TOEL	TU	PMSD
Angular (Corrected)	C > T	100	>100	n/a	1	16.81%

Dunnett Multiple Comparison Test

Control	vs	Conc-%	Test Stat	Critical	MSD	DF	P-Type	P-Value	Decision(α:5%)
Lab Water		13	0.3627	2.407	0.235	6	CDF	0.7053	Non-Significant Effect
		25	0.6593	2.407	0.235	6	CDF	0.5766	Non-Significant Effect
		50	0.3627	2.407	0.235	6	CDF	0.7053	Non-Significant Effect
		76	-0.05387	2.407	0.235	6	CDF	0.8487	Non-Significant Effect
		100	-0.8332	2.407	0.235	6	CDF	0.9732	Non-Significant Effect

Test Acceptability Criteria

Attribute	Test Stat	Lower	Upper	Overlap	Decision
Control Resp	0.95	0.8	>>	Yes	Passes Criteria

ANOVA Table

Source	Sum Squares	Mean Square	DF	F Stat	P-Value	Decision(α:5%)
Between	0.0517854	0.0103571	5	0.5415	0.7425	Non-Significant Effect
Error	0.344306	0.0191281	18			
Total	0.396091		23			

Distributional Tests

Attribute	Test	Test Stat	Critical	P-Value	Decision(α:1%)
Variances	Levene Equality of Variance Test	2.671	4.248	0.0563	Equal Variances
Variances	Mod Levene Equality of Variance Test	1.245	4.248	0.3297	Equal Variances
Distribution	Shapiro-Wilk W Normality Test	0.9004	0.884	0.0219	Normal Distribution

7d Survival Rate Summary

Conc-%	Code	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	L	4	0.9500	0.8581	1.0000	0.9500	0.9000	1.0000	0.0289	6.08%	0.00%
13		4	0.9250	0.7727	1.0000	0.9500	0.8000	1.0000	0.0479	10.35%	2.63%
25		4	0.9000	0.6750	1.0000	0.9500	0.7000	1.0000	0.0707	15.71%	5.26%
50		4	0.9250	0.7727	1.0000	0.9500	0.8000	1.0000	0.0479	10.35%	2.63%
76		4	0.9500	0.7909	1.0000	1.0000	0.8000	1.0000	0.0500	10.53%	0.00%
100		4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	0.00%	-5.26%

Angular (Corrected) Transformed Summary

Conc-%	Code	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	L	4	1.331	1.181	1.48	1.331	1.249	1.412	0.04705	7.07%	0.00%
13		4	1.295	1.061	1.529	1.331	1.107	1.412	0.07348	11.35%	2.67%
25		4	1.266	0.9499	1.582	1.331	0.9912	1.412	0.09936	15.70%	4.85%
50		4	1.295	1.061	1.529	1.331	1.107	1.412	0.07348	11.35%	2.67%
76		4	1.336	1.093	1.578	1.412	1.107	1.412	0.07622	11.41%	-0.40%
100		4	1.412	1.412	1.412	1.412	1.412	1.412	0	0.00%	-6.12%

CETIS Analytical Report

Report Date: 30 Dec-19 11:40 (p 2 of 2)
Test Code/ID: 5B3693B4 / 15-3030-3412

Fathead Minnow 7-d Larval Survival and Growth Test

ERM

Analysis ID: 00-1513-8377 Endpoint: 7d Survival Rate CETIS Version: CETISv1.9.4
Analyzed: 30 Dec-19 11:40 Analysis: Parametric-Control vs Treatments Status Level: 1

7d Survival Rate Detail

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4
0	L	0.9000	0.9000	1.0000	1.0000
13		1.0000	0.8000	1.0000	0.9000
25		1.0000	0.7000	0.9000	1.0000
50		0.9000	1.0000	0.8000	1.0000
76		1.0000	1.0000	0.8000	1.0000
100		1.0000	1.0000	1.0000	1.0000

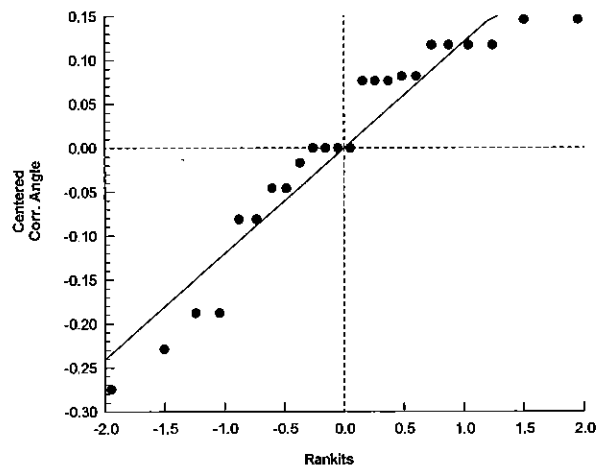
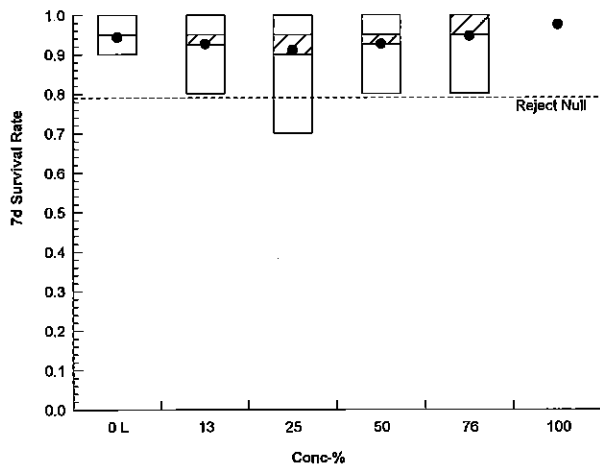
Angular (Corrected) Transformed Detail

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4
0	L	1.249	1.249	1.412	1.412
13		1.412	1.107	1.412	1.249
25		1.412	0.9912	1.249	1.412
50		1.249	1.412	1.107	1.412
76		1.412	1.412	1.107	1.412
100		1.412	1.412	1.412	1.412

7d Survival Rate Binomials

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4
0	L	9/10	9/10	10/10	10/10
13		10/10	8/10	10/10	9/10
25		10/10	7/10	9/10	10/10
50		9/10	10/10	8/10	10/10
76		10/10	10/10	8/10	10/10
100		10/10	10/10	10/10	10/10

Graphics



CETIS Analytical Report

 Report Date: 30 Dec-19 11:41 (p 1 of 2)
 Test Code/ID: 5B3693B4 / 15-3030-3412

Fathead Minnow 7-d Larval Survival and Growth Test

ERM

Analysis ID: 18-6383-0847	Endpoint: Mean Dry Biomass-mg	CETIS Version: CETISv1.9.4
Analyzed: 30 Dec-19 11:40	Analysis: Nonparametric-Control vs Treatments	Status Level: 1
Batch ID: 10-5585-1802	Test Type: Growth-Survival (7d)	Analyst: Lab Tech
Start Date: 17 Dec-19 16:00	Protocol: EPA/821/R-02-013 (2002)	Diluent: Reconstituted Water
Ending Date: 24 Dec-19 15:00	Species: Pimephales promelas	Brine:
Test Length: 6d 23h	Taxon: Actinopterygii	Source: In-House Culture Age: <24
Sample ID: 06-4191-1844	Code: 2642CC24	Project: WET Compliance Testing
Sample Date: 16 Dec-19 07:00	Material: POTW Effluent	Source: City of Cadillac WWTP
Receipt Date: 17 Dec-19 11:30	CAS (PC):	Station: Outfall 001A
Sample Age: 33h (1 °C)	Client: City of Cadillac WWTP	

Data Transform	Alt Hyp	NOEL	LOEL	TOEL	TU	PMSD
Untransformed	C > T	100	>100	n/a	40	34.15%

Steel Many-One Rank Sum Test

Per MDSGLS

Control	vs	Conc-%	Test Stat	Critical	Ties	DF	P-Type	P-Value	Decision(α:5%)
Lab Water		13	20	10	0	6	Asymp	0.9516	Non-Significant Effect
		25	19.5	10	1	6	Asymp	0.9315	Non-Significant Effect
		50	23	10	0	6	Asymp	0.9966	Non-Significant Effect
		76	23	10	0	6	Asymp	0.9966	Non-Significant Effect
		100	22	10	0	6	Asymp	0.9908	Non-Significant Effect

Test Acceptability Criteria

TAC Limits

Attribute	Test Stat	Lower	Upper	Overlap	Decision
Control Resp	0.491	0.25	>>	Yes	Passes Criteria

ANOVA Table

Source	Sum Squares	Mean Square	DF	F Stat	P-Value	Decision(α:5%)
Between	0.0218078	0.0043616	5	0.4495	0.8081	Non-Significant Effect
Error	0.174644	0.0097024	18			
Total	0.196451		23			

Distributional Tests

Attribute	Test	Test Stat	Critical	P-Value	Decision(α:1%)
Variances	Bartlett Equality of Variance Test	25.94	15.09	9.2E-05	Unequal Variances
Distribution	Shapiro-Wilk W Normality Test	0.7317	0.884	2.7E-05	Non-Normal Distribution

Mean Dry Biomass-mg Summary

Conc-%	Code	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	L	4	0.491	0.4162	0.5658	0.4805	0.446	0.557	0.0235	9.57%	0.00%
13		4	0.4732	0.1097	0.8368	0.5145	0.159	0.705	0.1142	48.28%	3.62%
25		4	0.5038	0.4524	0.5551	0.501	0.471	0.542	0.01615	6.41%	-2.60%
50		4	0.5565	0.494	0.619	0.554	0.511	0.607	0.01965	7.06%	-13.34%
76		4	0.5475	0.4988	0.5962	0.5435	0.516	0.587	0.01529	5.59%	-11.51%
100		4	0.531	0.5046	0.5574	0.532	0.511	0.549	0.008286	3.12%	-8.15%

Mean Dry Biomass-mg Detail

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4
0	L	0.446	0.557	0.484	0.477
13		0.495	0.705	0.159	0.534
25		0.471	0.484	0.542	0.518
50		0.553	0.607	0.555	0.511
76		0.554	0.587	0.533	0.516
100		0.525	0.511	0.549	0.539

CETIS Analytical Report

Report Date: 30 Dec-19 11:41 (p 2 of 2)
Test Code/ID: 5B3693B4 / 15-3030-3412

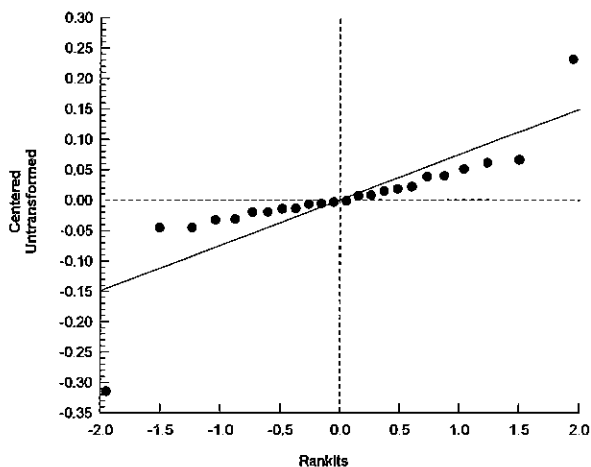
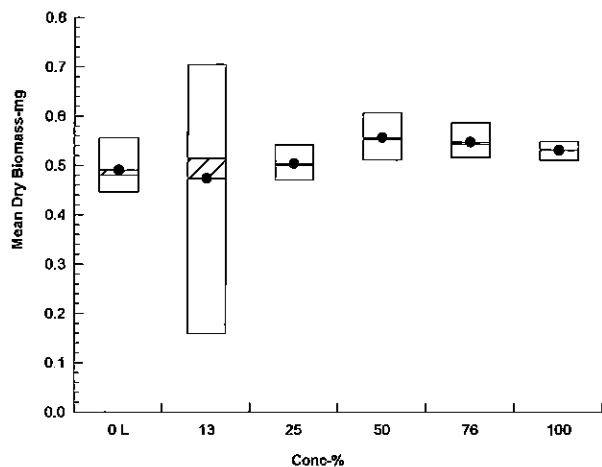
Fathead Minnow 7-d Larval Survival and Growth Test

ERM

Analysis ID: 18-6383-0847
Analyzed: 30 Dec-19 11:40
Endpoint: Mean Dry Biomass-mg
Analysis: Nonparametric-Control vs Treatments

CETIS Version: CETISv1.9.4
Status Level: 1

Graphics





3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

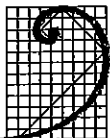
ERM[®]

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	CITY OF CADILLAC			SAMPLER													
ADDRESS:	1211 PLETT RD CADILLAC MI			PHONE NUMBER:													
SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)										
Tert.	12-15-19	7:05AM			pH= 7.18 s.u. NH ₃ = 0.12 mg/L	121719-5	Temp. 10.0 °C □ On Ice	D.O. 9.6 mg/L	pH 6.8 s.u.	Cond 910	umhos/cm						
	12-16-19	7:00AM	comp.	3 gal.	pH= s.u. NH ₃ = mg/L		Temp. °C □ On Ice	D.O. mg/L	pH s.u.	Cond	umhos/cm						
					pH= s.u. NH ₃ = mg/L		Temp. °C □ On Ice	D.O. mg/L	pH s.u.	Cond	umhos/cm						
					pH= s.u. NH ₃ = mg/L		Temp. °C □ On Ice	D.O. mg/L	pH s.u.	Cond	umhos/cm						
					pH= s.u. NH ₃ = mg/L		Temp. °C □ On Ice	D.O. mg/L	pH s.u.	Cond	umhos/cm						
					pH= s.u. NH ₃ = mg/L		Temp. °C □ On Ice	D.O. mg/L	pH s.u.	Cond	umhos/cm						
					pH= s.u. NH ₃ = mg/L		Temp. °C □ On Ice	D.O. mg/L	pH s.u.	Cond	umhos/cm						
ANALYSES REQUESTED [check item(s)]	Test Material: ☒ Water/Wastewater ☐ Sediment ☐ Product		Test Type: ☒ Acute ☒ Chronic ☐ Other		Test Species: ☒ Ceriodaphnia dubia ☐ Daphnia magna ☐ Daphnia pulex ☒ Fathead minnow (Pimephales promelas) ☐ Rainbow Trout (Oncorhynchus mykiss) ☐ Sheepshead minnow (Cyprinodon variegatus) ☐ Silverside minnow (Menidia beryllina) ☐ Other (write in comments section)												
COMMENT SECTION: ONLY NEED CHRONIC ON 2 SPECIES. THANKS																	
SAMPLE TRANSFERS																	
RELINQUISHED BY: Signature / Organization			DATE			TIME			ACCEPTED BY: Signature / Organization			DATE			TIME		
T. Lazjko / City of Cadillac			12-16-19			9:45AM			[Signature]			12/16/19			11:58		

* See Instructions for Sample Collection on Back of Sheet

February 2018



ERM®

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

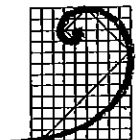
CLIENT NAME:	City of Cadillac				SAMPLER	TL				
ADDRESS:	1121 Platt Rd Cadillac, MI				PHONE NUMBER:	231-775-2841				
SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
Tert.	11-17-19	7:00 AM	Comp	3 gal.	pH=6.97 s.u. NH ₃ = .240 mg/L	12/9/19-2	Temp. 2 (°C) □ On Ice	D.O. 9.3 mg/L	pH 7.0 s.u.	Cond 1045 umhos/cm
	12-18-19	7:00 AM			pH= s.u. NH ₃ = mg/L		Temp. (°C) □ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L		Temp. (°C) □ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L		Temp. (°C) □ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L		Temp. (°C) □ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
ANALYSES REQUESTED [check item(s)]	Test Material: ☒ Water/Wastewater ____ Sediment ____ Product		Test Type: ☒ Acute ☒ Chronic ____ Other		Test Species: ☒ <i>Ceriodaphnia dubia</i> _____ Rainbow Trout (<i>Oncorhynchus mykiss</i>) ____ <i>Daphnia magna</i> _____ Sheephead minnow (<i>Cyprinodon variegatus</i>) ____ <i>Daphnia pulex</i> _____ Silverside minnow (<i>Menidia beryllina</i>) ☒ Fathead minnow (<i>Pimephales promelas</i>) _____ Other (write in comments section)					
COMMENT SECTION: ONLY CHRONIC - ON TWO SPECIES. AT 12/10/19										

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
T. Saegheh / City of Cadillac	12-18-19		Tonyan Bishop / ERM	12/19/19	1100

* See Instructions for Sample Collection on Back of Sheet

February 2018



ERM®

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	CITY OF CADILLAC				SAMPLER	CHAD M.				
ADDRESS:	1121 PLETT RD. CADILLAC				PHONE NUMBER:	231-775-2841				
SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
TEST	12-19	7:00 am	C	4-1 gal.	pH= 7.05 s.u. NH ₃ = 241 mg/L	122119-1	Temp. (°C) ☒ On Ice	D.O. 12 mg/L	pH 6.9 s.u.	Cond 1067 umhos/cm
	12-20	7:00 am			pH= s.u. NH ₃ = mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L		Temp. (°C) ☐ On Ice	D.O. mg/L	pH s.u.	Cond umhos/cm
ANALYSES REQUESTED [check item(s)]	Test Material: ☒ Water / Wastewater ☐ Sediment ☐ Product	Test Type: ☐ Acute ☒ Chronic ☐ Other	Test Species: ☒ <i>Ceriodaphnia dubia</i> — Rainbow Trout (<i>Oncorhynchus mykiss</i>) ☐ <i>Daphnia magna</i> — Sheepshead minnow (<i>Cyprinodon variegatus</i>) ☐ <i>Daphnia pulex</i> — Silverside minnow (<i>Menidia beryllina</i>) ☒ Fathead minnow (<i>Pimephales promelas</i>) — Other (write in comments section)							
COMMENT SECTION:										

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
Chad Waveguard	12-20-19	1:30 pm	Ship Next Day / Sat WPS		
			Kate Marking / ERM	12/21/19	11:30

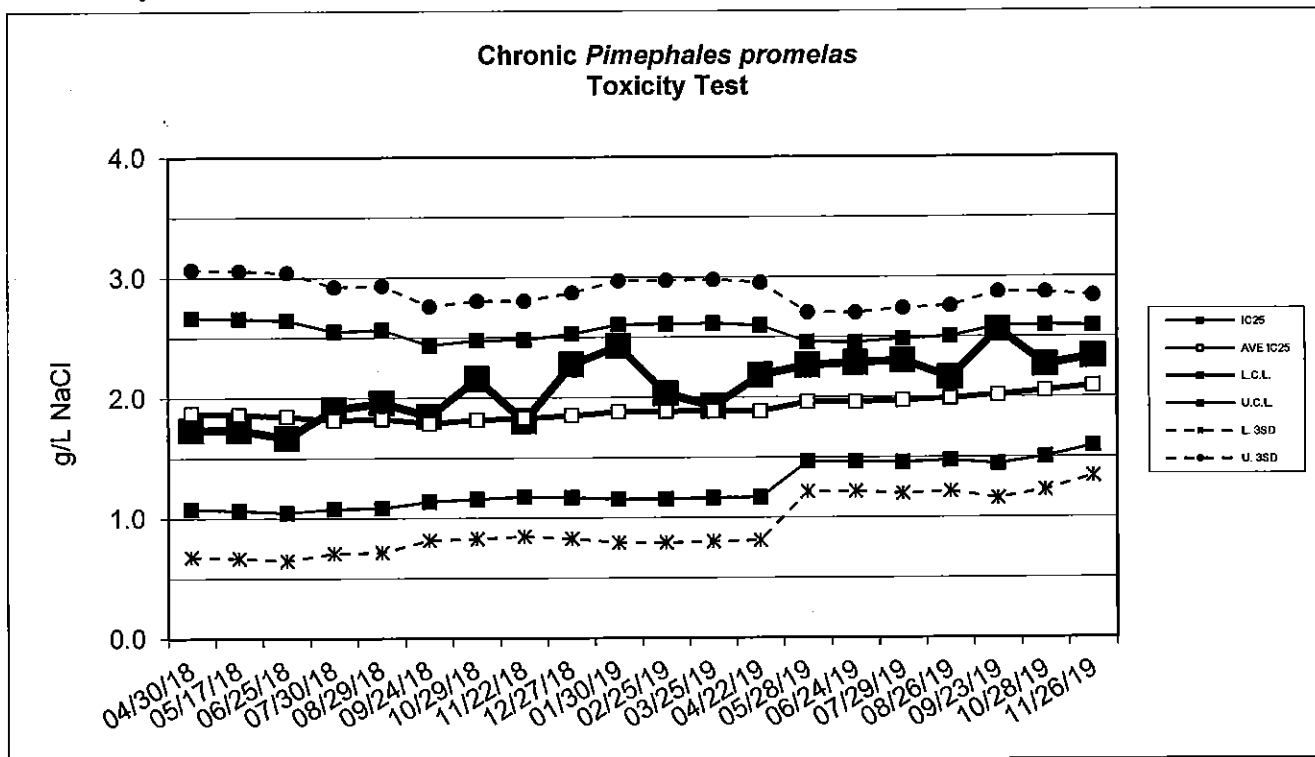
* See Instructions for Sample Collection on Back of Sheet

February 2018



Environmental Resources Management

Standard Reference Toxicant Data



Chronic *Pimephales promelas* Toxicity Test Data

Date	IC25 (g/L NaCl)	AVE IC25 (g/L NaCl)	CONTROL LIMIT		Survival (%)	CONTROL Growth (mg)	CV (%)
			Lower	Upper			
04/30/18	1.7	1.9	1.1	2.7	95	0.45	11.4
05/17/18	1.7	1.9	1.1	2.7	100	0.54	10.8
06/25/18	1.7	1.8	1.0	2.6	95	0.56	17.8
07/30/18	1.9	1.8	1.1	2.6	97.5	0.43	4.3
08/29/18	2.0	1.8	1.1	2.6	100	0.58	9.4
09/24/18	1.8	1.8	1.1	2.4	97.5	0.46	8.2
10/29/18	2.2	1.8	1.2	2.5	97.5	0.45	7.7
11/22/18	1.8	1.8	1.2	2.5	95	0.65	5.2
12/27/18	2.3	1.8	1.2	2.5	97.5	0.64	7.4
01/30/19	2.4	1.9	1.2	2.6	100	0.53	10.5
02/25/19	2.0	1.9	1.2	2.6	95	0.53	10.2
03/25/19	1.9	1.9	1.2	2.6	97.5	0.63	6.0
04/22/19	2.2	1.9	1.2	2.6	100	0.57	2.0
05/28/19	2.3	2.0	1.5	2.5	100	0.68	10.4
06/24/19	2.3	2.0	1.5	2.5	92.5	0.48	11.0
07/29/19	2.3	2.0	1.5	2.5	100	0.51	5.6
08/26/19	2.2	2.0	1.5	2.5	100	0.38	15.0
09/23/19	2.6	2.0	1.4	2.6	97.5	0.52	5.8
10/28/19	2.3	2.0	1.5	2.6	100	0.47	5.2
11/26/19	2.3	2.1	1.6	2.6	100	0.51	3.8

Appendix B
MDEGLE Reporting Form



MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

FATHEAD MINNOW CHRONIC TOXICITY TEST REPORT

By authority of PA 451 of 1994, as amended.

INSTRUCTIONS: Use this form to report chronic toxicity test results. Use separate forms for more than one test.

1. NAME OF FACILITY (on NPDES permit) City of Cadillac WWTP		2. NPDES PERMIT # M I 0 0 2 0 2 5 7						
1. RECEIVING WATER (as designated in permit) Clam River		2. OUTFALL 001A		5. RECEIVING WATER CONCENTRATION (if known)				
6. TEST LAB (Name and Address) Environmental Resources Management 3352 128 th Avenue Holland, MI 49424-9263								
7. TEST START DATE 12/17/19		8. TEST END DATE 12/24/19		9. AGE RANGE OF ORGANISMS AT TEST START <24 Hrs Old			10. REPORT DATE 01/07/20	
11. NAME OF PERSON CONDUCTING TEST Lab Contact: Mark Baker				12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT Bruce Rabe (616) 738 – 7308				
13. SAMPLE COLLECTION DATES Sample 1: 12/16/19 Sample 2: 12/18/19 Sample 3: 12/20/19		14. DATE RECEIVED Sample 1: 12/17/19 Sample 2: 12/19/19 Sample 3: 12/21/19		15. ARRIVAL TEMP (C°) Sample 1: 1 Sample 2: 2 Sample 3: 1				
16. DATE OF FIRST USE Sample 1: 12/17/19 Sample 2: 12/19/19 Sample 3: 12/21/19		17. TOTAL RESIDUAL CHLORINE (mg/l) Sample 1: <0.01 Sample 2: <0.01 Sample 3: <0.01		18. AMMONIA (mg/l as N) Sample 1: 0.14 Sample 2: 0.18 Sample 3: 0.12				
19. WAS SAMPLE DECHLORINATED? Sample 1: ☐ YES ☒ NO IF YES ➡ Sample 2: ☐ YES ☒ NO IF YES ➡ Sample 3: ☐ YES ☒ NO IF YES ➡		20. DESCRIBE DECHLORINATION (if any)						
21. EFFLUENT SAMPLES WERE COLLECTED (check one) ☐ BEFORE CHLORINATION ☐ AFTER CHLORINATION ☐ AFTER CHLORINATION, BEFORE DECHLORINATION ☐ AFTER DECHLORINATION ☒ FACILITY DOES NOT CHLORINATE								
22. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded holding time. Attach a separate sheet if necessary.) The percent minimum significant difference (MSDp) value of 34.2 percent for growth exceeded the maximum allowed value of 30 percent. An MSDp exceedance can mask statistically significant reductions in growth when comparing effluent responses to the control. However, based on the greater response for both survival and growth in the 100 percent effluent when compared to the control, the MSDp exceedance was deemed a non-issue.								
23. EFFLUENT FILTERED? ☐ YES ☒ NO IF YES ➡		24. STATE MESH SIZE OF FILTER (if filtered)						
25. EFFLUENT SAMPLE TYPE (check one type for each sample) Sample 1: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE Sample 2: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE Sample 3: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE							26. IDENTIFY THE DILUENT (O ₁) CONTROL RMHW IDENTIFY THE SECONDARY (O ₂) CONTROL (if used)	
27. SUMMARY OF DATA AND RESULTS - SURVIVAL AND GROWTH								
CONCENTRATION OF EFFLUENT (%)	O ₁ (dilution H ₂ O)	O ₂	13%	25%	50%	76%	100%	
96-HOUR SURVIVAL (%)	97.5	N/A	100	90	92.5	95	100	
7-DAY MEAN BIOMASS (mg/initial fish)	0.491	N/A	0.473	0.504	0.557	0.548	0.531	
7-DAY MEAN SURVIVAL (%)	95	N/A	92.5	90	92.5	95	100	
28. 96-HOUR LC ₅₀ (%) >100		29. TU _a (acute toxic units) 0						
30. 7-DAY CHRONIC VALUE (%) >100	31. NOEC 100%		32. LOEC >100%			33. TU _c (chronic toxic units) 0		

EQP 5820 (02/10)